MODULE NO. 20

DENSE SET:

Definition:

A subset M of a metric space X is said to be dense in X if $\bar{M} = X$

Closure set is a set along with its limits points.

Example (1):

The rational numbers $\mathbb{Q}$ are dense in $\mathbb{R}$.

we have infinite sets $\mathbb{Q}$ and $\mathbb{R}$,

Let $x \in \mathbb{R}$, and x can be a integer or fraction,

as $x = n + r$ *(e.g 2.123=2+0.123)*

$n \in \mathbb{Z}$, $0 \le r < 1$ if r is between 0 and 1 then it is fraction.

here, $r = 0.r_1 r_2 r_3 .....$

set $x_k = n + 0.r_1 r_2 r_3 .....r_k$

So, each x_k is a rational number as we fix the fractional part $r = 0.r_1 r_2 r_3 .....$ when we fix the fractional part then it gives you rational number. Real number may be rational(fraction part is fix and not continue) or irrational (fraction part is not fix and continue).

$$\lim_{x \to \infty} x_k = x$$

here x_k is a rational number and at $x \to \infty$ and it gives irrational number x. so all rational numbers cover all irrational numbers, $\bar{Q} = \mathbb{R}$ this shows that Q is dense in $\mathbb{R}$, as real set $\mathbb{R}$ contains rational and irrational numbers, if Q gives a rational number then it is also in Q and also in $\mathbb{R}$ but if Q gives an irrational number then it is also present in $\mathbb{R}$.

MODULE NO. 58

LINEAR OPERATORS

➢ *Theorem (Inverse Operator)*

Long

Theorem

Let X, Y be vectors spaces, both real or both complex. Let $T : D(T) \to Y$ be a linear operator with domain $D(T) \subset X$ and range $R(T) \subset Y$ .then:

Quiz

a): **The inverse $T' : R(T) \to D(T)$ exists if and only if $Tx=0 \Rightarrow x=0$. (i.e null space has zero elements).**

Quiz

b): **If T' exists, it is a linear operator.**

Quiz

c): **if $\dim D(T) = n < \infty$ *and* T^{-1} exists, then $\dim R(T) = \dim D(T)$.**

as there is if and only if condition so we have to prove in both ways.

MODULE NO. 52

FINITE DIMENSION

➢ Theorem (Finite Dimension)

Theorem

Quiz

If a normed space X has the property that the closed unit ball $M = \{x \mid \|x\| \leq 1\}$ is compact, then X is finite dimensional.

Proof Suppose on contrary that M is compact but $\dim X = \infty$,

Let $x_1 \in X$ s.t $\|x_1\| = 1$

it generates one dimensional subspace X_1 of X

$\Rightarrow$ finite dim $\Rightarrow$ compact $\Rightarrow$ closed. Since it is proper subspace of X_1, by Riesz's Lemma $\exists$ a $x_2 \in X$ with $\|x_2\| = 1$ s.t

$$\|x_2 - x_1\| \geq \theta = \tfrac{1}{2} \text{ (say)} \quad \theta \in (0,1)$$

Again x_1, x_2 generate a two dimensional proper closed subspace X_2 of X. Again by Riesz's Lemma $\exists$ $x_3 \in X$ s.t. $\|x_3\| = 1$ and $\forall\, x \in X_2$ we have

$$\|x_3 - x\| \geq \tfrac{1}{2}$$

in particular since $x_1, x_2 \in X_2$

$$\Rightarrow \|x_3 - x_1\| \geq \tfrac{1}{2}$$

$$\|x_3 - x_2\| \geq \tfrac{1}{2}$$

$y=(\eta_1,\eta_2,.....,\eta_n)$ where $\eta_i \in \mathbb{R}$

The distance function $d(x,y)=\|x-y\|$

$$d(x,y)=\sqrt{|\xi_1-\eta_1|^2+.....+|\xi_n-\eta_n|^2}$$

Unitary Space $\mathbb{C}^n$ is normed space?

Long

This is a metric space and elements in $\mathbb{C}^n$ is in n-tuples form,

$x=(\xi_1,\xi_2,.....,\xi_n)$ where $\xi_i \in \mathbb{C}$, $x \in X$

2) Prove that Real-line is metric space.

let x and y be two real points on real line, then

$$d(x,y) = |x-y| \qquad x, y \in \mathbb{R}$$

Now we Prove all the 4 axioms

$$d(x,y) = |x-y|$$

$$d(x,y) = |x-z+z-y|$$

$$d(x,y) \leq |x-z| + |z-y|$$

$$d(x,y) = d(x,z) + d(z,y)$$

Show that $d(x,y) = \sqrt{|x-y|}$ defines a metric on the set of all real number.

For $x, y, z \in X = R$, we need to verify the axioms of metric. (M1) to (M3) follows easily from properties of absolute value. To verify (M4), for any $\in R$ we have

$$[d(x,y)]^2 = |(x-y)| \leq |x-z| + |z-y|$$

$$\leq |x-z| + |z-y| + 2\sqrt{|x-z|}\sqrt{|z-y|}$$

$$= \left(\sqrt{|x-z|} + \sqrt{|z-y|}\right)^2$$

$$= \left[d(x,z) + d(z,y)\right]^2$$

Taking square root on both sides yields the triangle inequality.